

How RAG Works: Tokens, Embeddings, MongoDB & Qdrant

Concrete Technical Deep Dive using "The Wonderful Wizard of Oz" by L. Frank Baum

Step 1: Document Text & Tokenization

Raw Document Text (UTF-8 Byte Stream)

"Dorothy lived in the midst of the great Kansas prairies, with Uncle Henry, who was a farmer, and Aunt Em, who was the farmer's wife. Their house was small..."

Raw String: 198 characters • 34 words • UTF-8 Byte Stream

↓ Byte-Pair Encoding (BPE) Tokenizer ↓

Token Sub-Word Segmentation (Vocabulary IDs):

Dorothy #14820	lived #5829	in #287	the #262	midst #27192	of #286	the #262
great #1049	Kansas #14902	prairies #39108	farmer #8912	... (42 tokens)		

• 1 token ≈ 4 chars • Tokens map words to integer vectors

Step 2: Dense Semantic Embeddings

Embedding Model: nomic-embed-text (or text-embedding-004)

Transforms token sequence into a normalized 768-dim float32 vector

$v = [0.0412, -0.0891, 0.1205, \dots, 0.0074] \in \mathbb{R}^{768}$
Normalized length: $\|v\| = 1.0$ (Unit hypersphere for cosine similarity)

Semantic Hyperspace (Geometric Distance = Meaning Similarity):

Dimension 2 (Characters / Roles)



Cosine Similarity: $\cos(\theta) = (u \cdot v) / (\|u\| \|v\|) \in [-1.0, +1.0]$

Step 3: Dual Database Architecture

Raw Document Text (UTF-8 Byte Stream)

Collection: "documents" • Full Record & Version History

```
{
  "_id": ObjectId("66fc10..."),
  "title": "The Wonderful Wizard of Oz",
  "bibtex": { "author": "Baum, L. Frank", "year": "1900" },
  "fullText": "Dorothy lived in the midst of...",
  "version": 1, "summaries": { "llama": {...}, "mistral": {...} }
}
```

Qdrant Vector DB (HNSW Index Engine)

Collection: "document_chunks" • Sub-millisecond ANN

```
{
  "id": "c1b48f92-7f31-4a9c-9c71-09ba192f1b0a",
  "vector": [ 0.0412, -0.0891, 0.1205, ... ],
  "payload": {
    "documentId": "doc-oz-1900", "chunkIndex": 0,
    "text": "Dorothy lived in the midst of the great Kansas prairies..."
  }
}
```

Step 4: Interactive RAG Query Execution Pipeline (Retrieval → Augmentation → Generation)

Raw User Query (UTF-8 Byte Stream)

"Where did Dorothy live and what was her home like?"

• Query Tokens: 11 tokens
• Model: nomic-embed-text
• Vector: $v_{query} \in \mathbb{R}^{768}$



2. Qdrant HNSW Search

Filter: documentId = "doc-oz-1900"
Metric: Cosine Similarity • Top-K = 1

Retrieved Document Snippet (UTF-8 Byte Stream)

"Dorothy lived in the midst of the great Kansas prairies... house was small"
Relevance: 89.2% • Latency: 2.1 ms



Raw Prompt Construction (UTF-8 Byte Stream)

[SYSTEM: Answer based strictly on context]
[CONTEXT CHUNK 0 (Relevance: 89%)]:
"Dorothy lived in Kansas prairies... four walls, a floor and a roof, one room..."
[USER QUESTION]:
"Where did Dorothy live and what was her home like?"



Raw LLM Output (UTF-8 Byte Stream)

Grounded Answer Output:
"Dorothy lived in the midst of the Kansas prairies with Uncle Henry and Aunt Em. Her home was a small, one-room wooden house..."

Final Answer String (UTF-8 Byte Stream)

Core Architectural Distinctions: MongoDB (Document Store) vs. Qdrant (Vector Database)

CAPABILITY / ROLE	MONGODB 7.0 (DOCUMENT REPOSITORY)	QDRANT (VECTOR RETRIEVAL ENGINE)
Primary Data Unit	Hierarchical BSON documents (full text, metadata, authors, version history)	High-dimensional points (vector float32 arrays + small payload metadata)
Query Mechanism	B-Tree index lookups, exact key matches, regex, aggregation pipelines	Approximate Nearest Neighbor (ANN) search via HNSW graphs (Cosine / Dot product)
RAG System Purpose	Single source of truth for raw assets, BibTeX fields, versions, and generated summaries	Sub-millisecond semantic passage locator to extract exact relevant context for prompt injection